

Samuel Chagas de Assis

COMPUTATIONAL BIOLOGIST RESEARCHER

Campinas, São Paulo - Brazil

✉ samuel.assis@lnbio.cnpem.br | 🏠 chagas98.github.io | 📷 chagas98 | 🌐 samuelchagass

About me

Master's student in Computational Biology and Pharmaceutical Sciences at the Brazilian Center for Research in Energy and Materials (CNPEM) and the University of Campinas (UNICAMP), currently conducting research with the Protein Design Group. My work focuses on applying and developing computational methods grounded in geometric deep learning to investigate protein-protein interactions. I am developing classification models using public immunological datasets, exploring graph-based representations and architectures, and leveraging model interpretability to uncover mechanistic insights into molecular recognition.

Education

Brazilian Center for Research in Energy and Materials (CNPEM) | State University of Campinas (UNICAMP)

Campinas/São Paulo, Brazil

MSC IN COMPUTATIONAL BIOLOGY AND PHARMACEUTICAL SCIENCES

2025-Present

Federal University of Latin-American Integration (UNILA)

Foz do Iguaçu, Brazil

BSC IN BIOTECHNOLOGY

2018-2024

Experience

Brazilian Center for Research in Energy and Materials (CNPEM)

Campinas, São Paulo

MSC. RESEARCH - GEOMETRIC DEEP LEARNING FOR PROTEINS

Feb 2025 - Present

- Developing graph-based deep learning models for protein-protein interaction prediction. Integrating PyTorch, PyTorch Geometric, and transformer-based embeddings to enhance protein structure representation.

Brazilian Center for Research in Energy and Materials (CNPEM)

Campinas, São Paulo

COMPUTATIONAL BIOCATALYSIS INTERN

Feb 2023 - Dec 2024

- I performed several molecular dynamics simulations and large-scale data analysis on HPC environment. The main goal are new insights about molecular mechanisms from new Carbohydrate-Active enzymes, which benefits the experimental team for mutation screening and activity assays.

SporeData Inc.

Remote

FREELANCER - JR DATA ANALYST

Oct 2022 - Feb 2023

- Worked as a freelancer on healthcare data projects, performing ETL, cohort generation, data visualization, clinical study reports, and statistical analyses.

Medical Genetics Lab - UNILA

Foz do Iguaçu, Brazil

UNDERGRADUATE RESEARCHER

Aug 2020 - Mar 2023

- I conducted computational analyses integrating DNA sequencing and HLA-B genotyping data from COVID-19 patients in Foz do Iguaçu. Using bioinformatics, I analyzed SARS-CoV-2 genomic variants and predicted HLA-epitope binding affinities to assess their potential impact on immune recognition and viral escape mechanisms

International Genetically Engineered Machine Competition (iGEM)

Paris, France

DRYLAB COORDINATOR

Jan 2020 - Dez 2021

- As the Head of DryLab at the iGEM UNILA LatAm Team, I led the development of a kinetic model with parameter optimization. We used an Ordinary Differential Equations (ODEs) framework coupled with sensitivity analysis to evaluate and optimized the performance of our synthetic genetic circuit.

Skills

Programming

PYTHON | BASH | R

Frameworks

PYTORCH | PLLM | NUMPY | SCIKIT-LEARN |

Workflow

NEXTFLOW

Container

DOCKER | SINGULARITY

Tools

GIT | GITHUB | SLURM | R/SHINY | PYTHON/FLASK

Accomplishments

- 2022 1st Place. Honorable Mention in Health Science at XI EICTI, UNILA
- 2021 Gold Medal at International Genetically Engineering Machine Competition - iGEM
- 2020 Best Public Choice Project at the V Biological Machines Engineering Summer Program - UFMG
- 2020 Best Presentation Award at the V Biological Machine Engineering Summer Program - UFMG
- 2019 1st Place. Honorable Mention in Technology and Production at VII SEUNI - UNILA